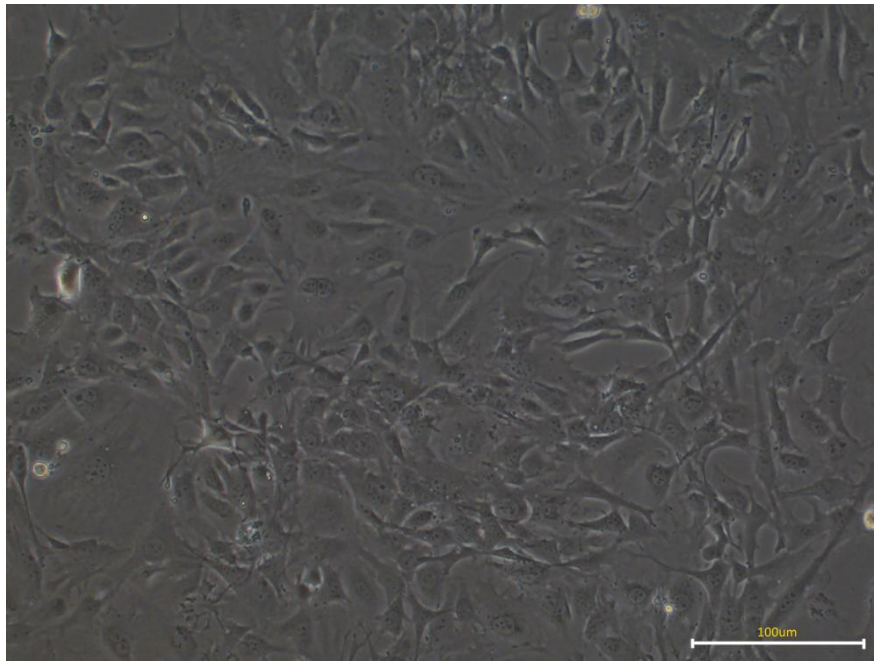
Stem Cell Biology

SECTION 3

MOUSE EMBRYONIC STEM CELLS (MESCS)



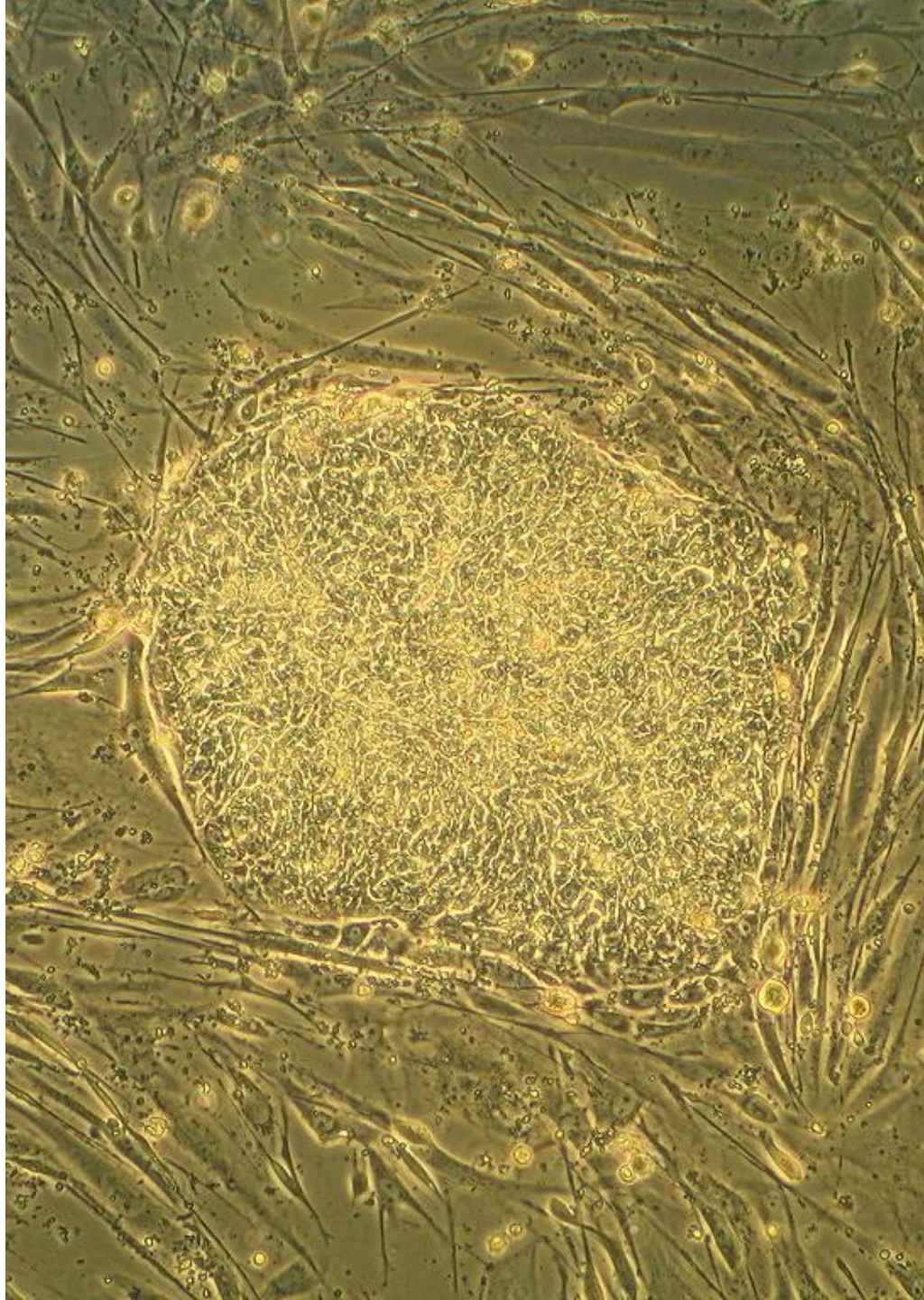
Mouse Embryonic Fibroblasts (MEFs)



- Mouse Embryonic Fibroblasts (MEFs) are a type of fibroblast prepared from mouse embryo.
- MEFs show a spindle shape when cultured in vitro, a typical feature of fibroblasts.

MEFs Application

- MEFs treated by mitomycin or gamma rays (such treatment makes MEF stop mitosis) are widely used as feeder in embryonic stem cell culture because they can mimic the microenvironment in embryo.



Materials

Reagents

Phosphate-buffered saline (PBS), without calcium or magnesium
70% (v/v) ethanol
MEF culture medium: D-MEM supplemented with 10% FBS, penicillin (100 U ml–1)/streptomycin (100 mg ml–1) and 2 mM L-glutamine.
Mitomycin C.

Reagents

FBS-ES medium, a medium similar to SR-ES medium, in which SR was replaced by 20% ES cell-pretested FBS.

0.25% trypsin-EDTA, 2.5% trypsin.

Embryonic day 12.5 (E12.5) or E13.5 mouse embryos

0.1% gelatin: dissolve 1 g of gelatin in MilliQ column–purified water and sterilize by autoclaving 30 min at 120 1C .

HEPES/DMEM, obtained by mixing Knockout D-MEM medium with HEPES buffer solution.

SR-ES medium

Equipments

35-, 60- and 100-mm tissue culture plates
12-well plates (Falcon, multi-well 12-well, cat. no. 353043)
10 ml pipette tips (Gilson, Diamond D10, cat. no. F161632;
20 ml pipette .23-G needles (0.6 * 25 mm) .1-ml syringes
Dissection forceps, Dissection scope and scissors
Inverted microscope with phase contrast

Procedure



Dissect out the uterine horns

1. *Briefly rinse in 70 % ethanol*
2. *Place into a falcon with PBS*

Separate each embryo from placenta and embryonic sac

Dissect head and red organs

1. *Wash in PBS*
2. *Place all embryos in a clean Petri dish*

Finely mince the tissue using a sterile razor blade until it becomes "pipettable"

1. *Add trypsin and DNase I*
2. *Incubate for 15 minutes at 37 °C*
3. *Pipet up and down after each 5 minutes incubation*
4. *Inactivate trypsin by adding MEF medium*
5. *Centrifuge at 300 x g, 5 minutes*
6. *Remove the supernatant*
7. *Resuspend cell pellet in MEF medium*

Plate 3-4 embryos in each T150 gelatine-coated flask

Freeze P0 | **Expand until P3 or P4**

1. Mitomycin C treatment

Feeder cells | **Conditioned media**

ELISA (Activin A)

Preparation of mouse embryonic fibroblasts (MEFs) (TIMING 2–3 d)

- **1|** Prepare and label 100-mm tissue culture plates.
- **2|** Kill pregnant female mouse between 12.5 and 13.5 days post coitum.
- **3|** Dissect out the uterine horns with embryos and briefly rinse them in 70% ethanol.
- Place uterine horns into a Petri dish and separate each embryo from its placenta and embryonic sac

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- **4|** Cut out the heads and remove internal abdominal and thoracic organs.
- Add 1 ml of 0.05% trypsin/EDTA (Gibco, Invitrogen), including 100 Kunitz units of DNase I (USB), per embryo.
- Trypsinize the tissue 2–4 min in the 37 1C incubator until tissues start to disintegrate.
- Quickly pipette all the tissue up and down with a 1-ml pipette to achieve efficient dissociation and proceed to Step 5.
- **5|** Inactivate the trypsin by adding about 1 volume of freshly prepared MEF medium.

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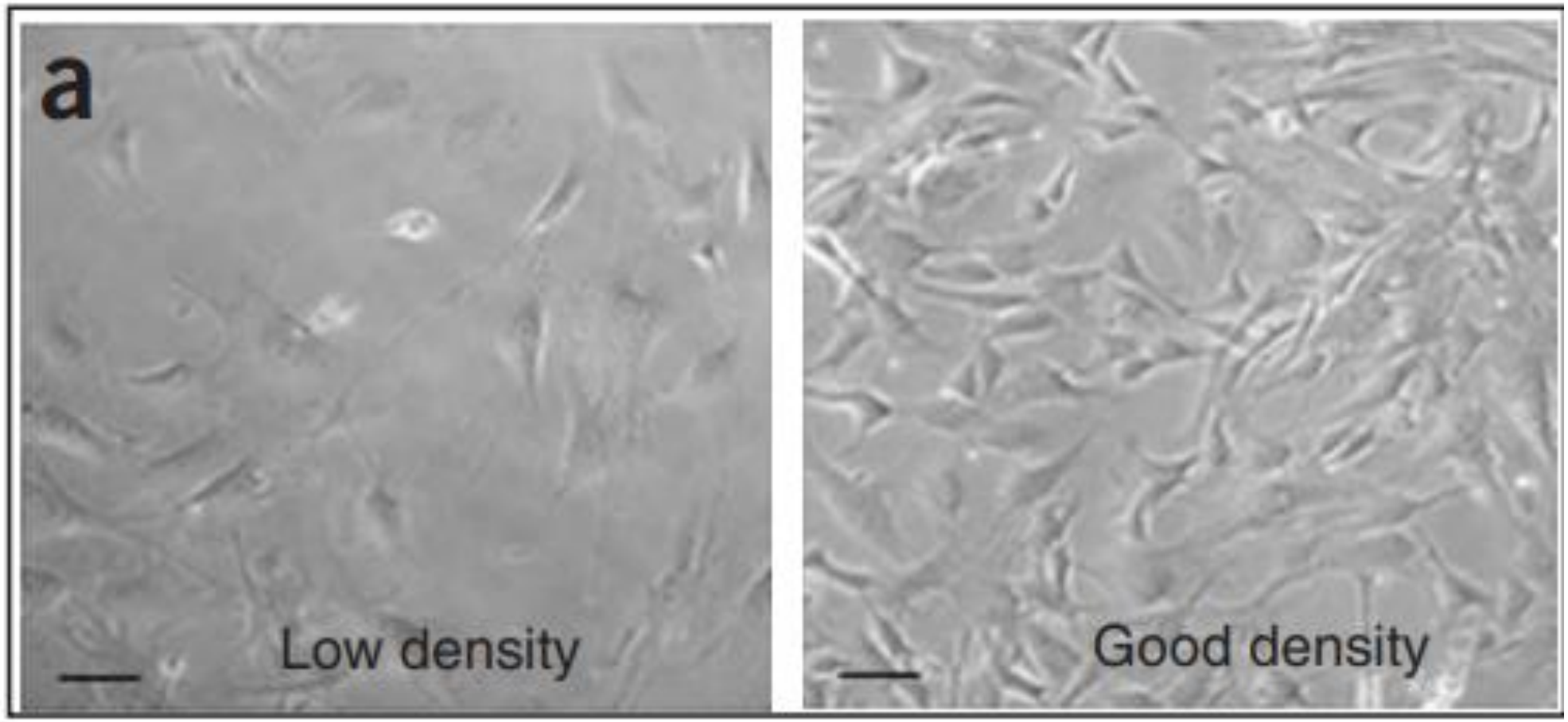
- **6|** Centrifuge and Collect the supernatant and seed out a homogenous suspension, keeping a ratio of two embryos per 100-mm plate.
- **7|** Change culture medium 4–5 h after seeding to remove unattached cells, and let the MEFs grow to 90% confluence (it usually takes 2–3 d). MEFs at this stage ready to be frozen are considered to be passage 1.
- **8|** Proceed to the next step, or freeze the MEFs down (cells from one 100-mm plate per single vial) using standard cell culture techniques.

Preparation of MEF feeder layer, Inactivation (TIMING 1 d)

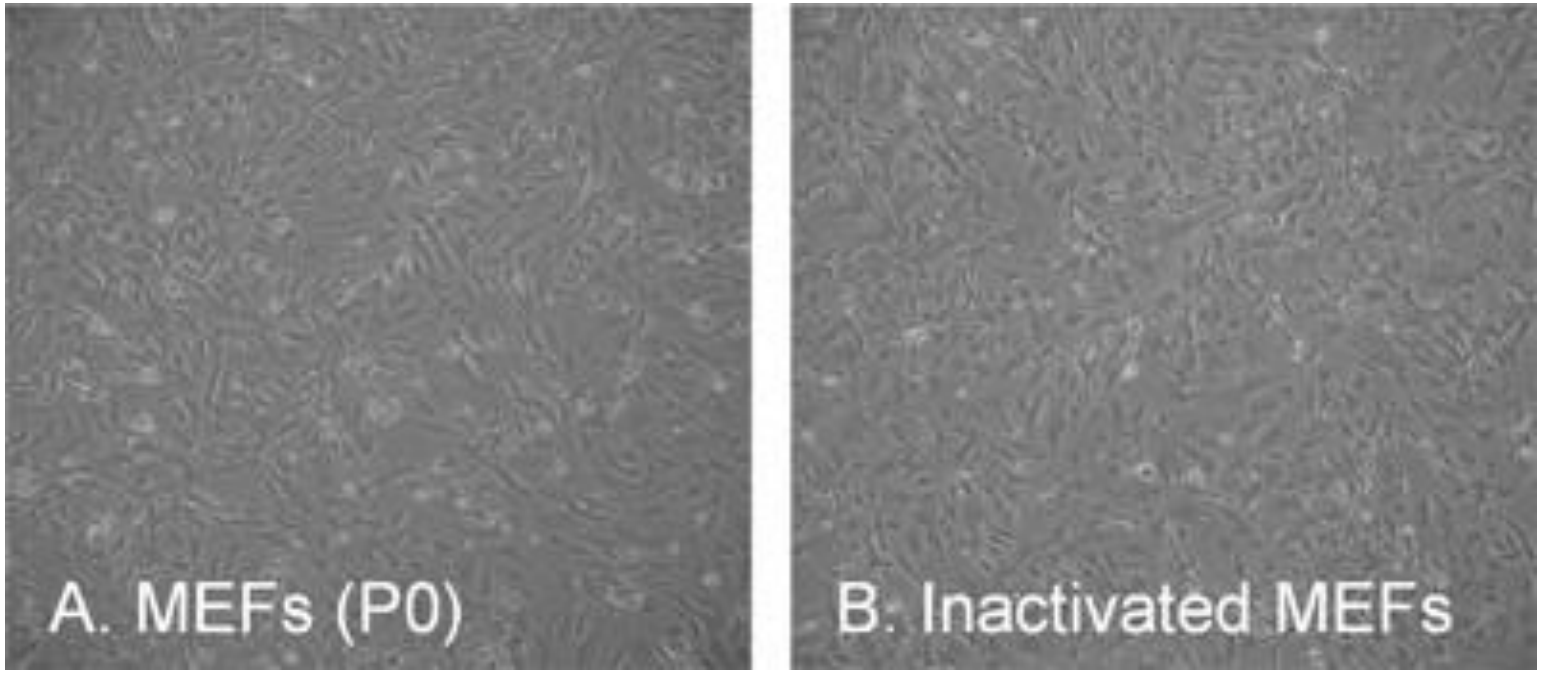
- **9|** Treat confluent MEFs with mitomycin C for 2 h (10 mg ml–1). Alternatively, cells may also be mitotically inactivated by exposure to 8,000 rad of X-ray irradiation.
- **10|** Coat T150 flasks with 0.2% gelatine and incubate at RT for at least 2 hr
- Remove the excess gelatin solution just before next step.

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- **11|** After mitomycin C inactivation, wash twice with PBS, trypsinize, centrifuge (for 5 min at 300 x g) and resuspend cells in warm medium.
- The feeder layer should cover the entire well (Fig. 1). Normally confluent MEFs from a 100-mm plate are sufficient for two entire confluent 12-well plates.



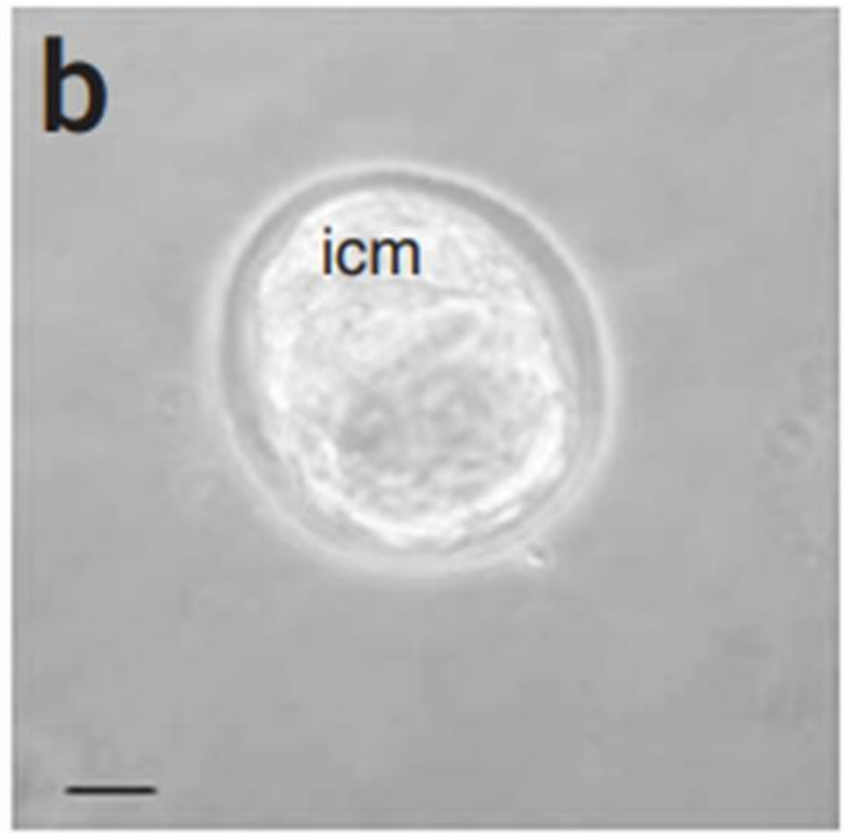
MEF feeder layer at low (insufficient) and proper density



The typical morphology of mouse embryonic fibroblasts (MEFs).
(A) Passage 0 (P0) two days after plating/isolation, (B) inactivated feeder layer at a density of 56.000 cells/cm.^{2}

Blastocyst extraction and ICM expansion (TIMING 5–6 d)

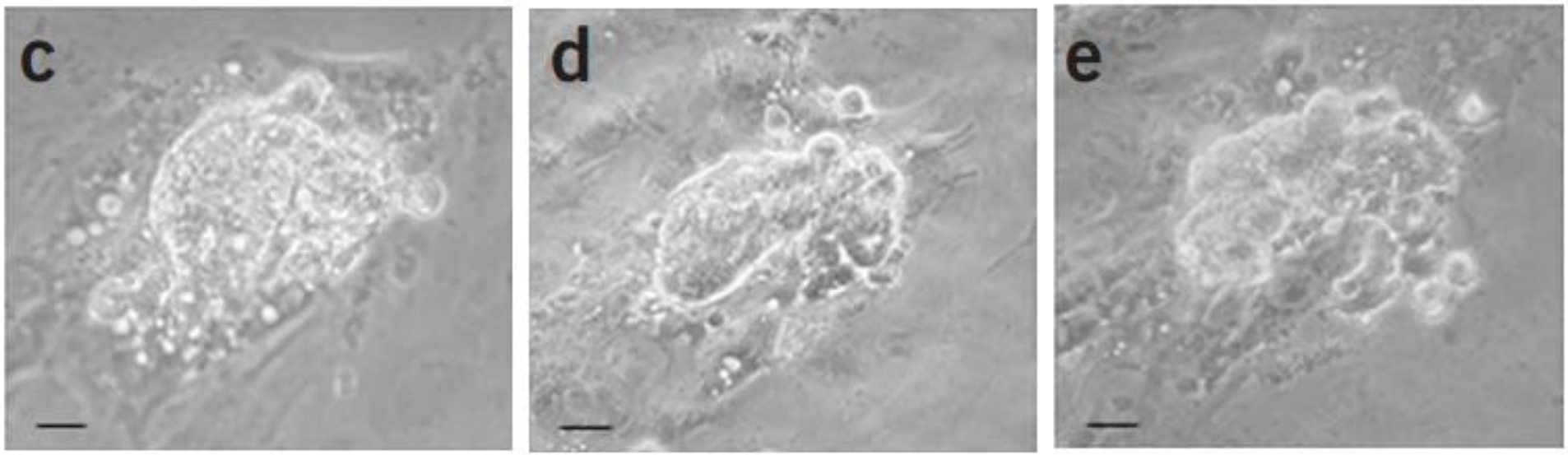
- **12|** Replace MEF medium with SR-ES medium 1–3 blastocysts extraction.
- **13|** Transfer the uteri into 2 ml of fresh preheated HEPES/DMEM in 35-mm dish.
- **14|** Flush the blastocysts out of the uterine horn under a dissection scope using a 1-ml syringe with 23-G needle.
- Collect blastocysts (Fig. 1b) under an inverted microscope using 10 ml pipette tips and a 20-ml pipette and transfer individually to wells of a 12-well plate containing a feeder layer of MEFs mitotically inactivated 1 d earlier.



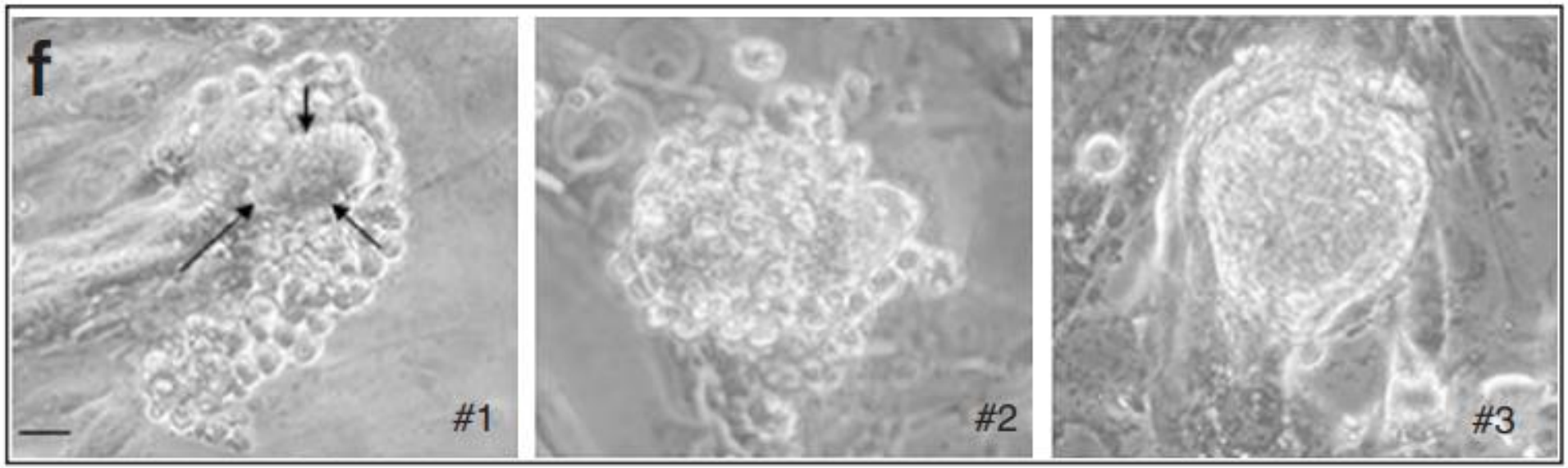
Morphology of mouse E3.5 blastocyst with inner cell mass (icm) indicated.

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- **16|** Allow blastocysts to attach to the MEF feeder layer. That usually takes 2 d. Avoid any interference during this period.
- After attachment, the blastocyst hatches (Fig. 1c).
- SR-ES medium can be then changed, and should subsequently be replaced every second day while the inner cell mass expands (Fig. 1d,e), until it adopts a morphology comparable to that shown in Figure 1f (usually on day 5 or 6).



(c–e) Blastocyst hatched on MEF feeder
2d (c), 3 d (d) and 4 d (e) after extraction.



(f) Examples of hatched blastocysts with expanded ICM and defined ESC like population (arrows), ready for the first trypsinization at day 5 or 6 after extraction.

First trypsinization and ESC expansion (TIMING 10–15 d)

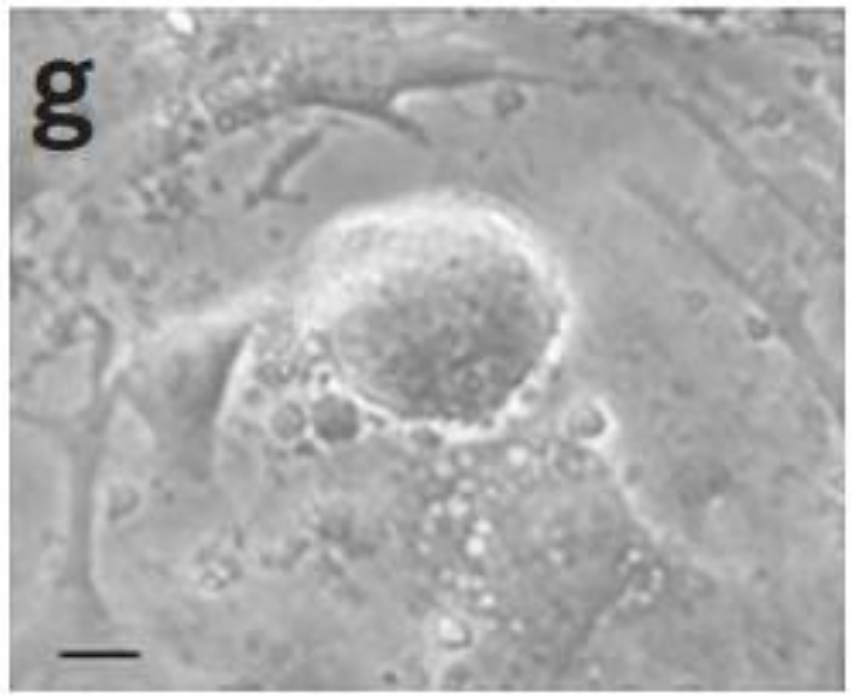
- **17|** Prepare and label appropriate numbers of 12-well plates with confluent layers of mitotically inactivated MEFs 1 d in advance.
- **18|** Two hours before trypsinization, replace the MEF medium with 1 ml per well of FBS-ES medium.
- **19|** Aspirate SR-ES medium from the wells containing expanded blastocysts on MEFs and add 200 ml of 0.25% trypsin-EDTA to the well.

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- **20|** After 15 s, aspirate the trypsin-EDTA, leaving only a wet surface.
- This procedure reduces stickiness of the cells and allows easier detachment. Proceed immediately to the next step.
- **21|** To mechanically detach an expanded blastocyst and surrounding MEFs, use a 20-ml pipette and 10-ml pipette tip.

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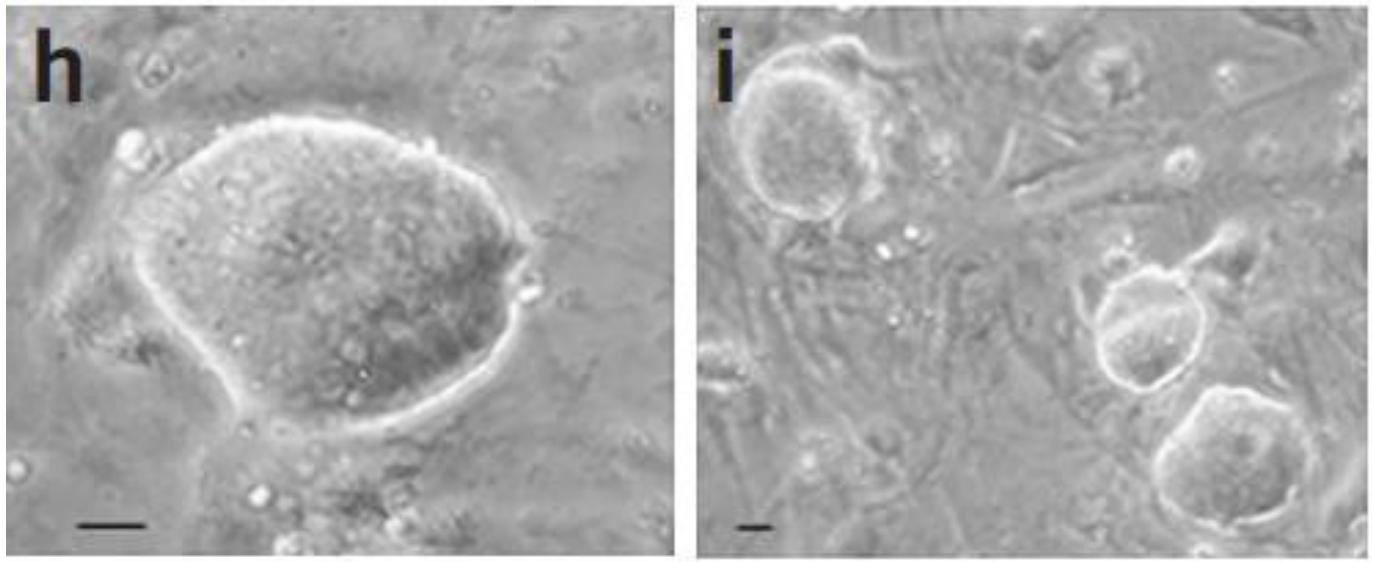
- **22|** Trypsinize the cells in a drop in the 37 1C incubator and then dissociate into individual cells and small cell clumps, by pipetting up and down approximately ten times with a 10-ml pipette tip.
- **23|** Several days after trypsinization, compact cell colonies with typical ESC colony morphology (Fig. 1g) will be detected in some of the wells (the number of colonies usually varies from 1 to 20).



Morphology of mESC-like colony 2 d after the first trypsinization.

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- **24|** Passage ESCs according to their needs. Trypsinize all cells in a well with 0.25% trypsin-EDTA when most colonies in the well reach the size and morphology shown in Figure 1h.
- Usually an additional trypsinization step is needed before the cells can be grown on 60-mm-diameter dishes to sub-confluency before being frozen as described Figure 1i.



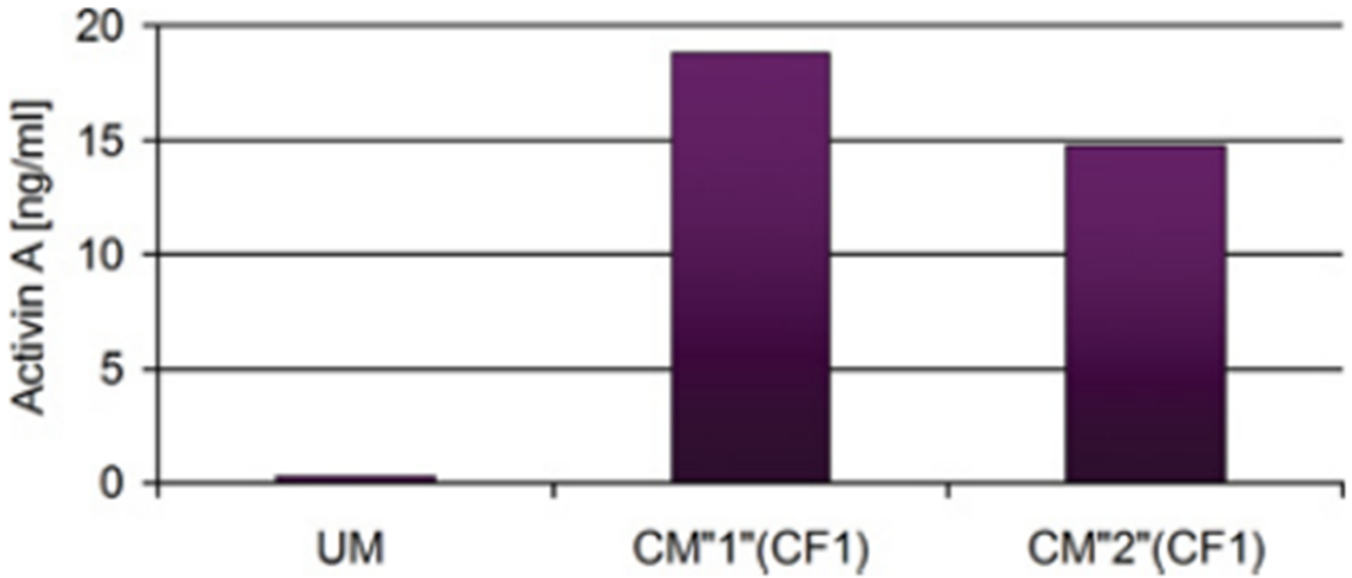
(h) Typical appearance of mESC colony at the stage when it should be subjected to the second trypsinization (usually 4–5 d after the first trypsinization).
(i) Established mESC line

Conditioned Medium (CM) Preparation

- **1.** The day after plating inactivated MEFs at a density of 56.000 cells/cm2, replace the MEF medium with hESC medium (unconditioned medium) supplemented freshly with 4 ng/ml of FGF2.
- **2.** Collect CM from feeder flasks after 24 h incubation and add fresh hESC medium containing 4 ng/ml of FGF2 to the feeders.
- **3.** Repeat this procedure for the next 6 days. Each day store collected CM at -20 °C.
- **4.** After 6 days mix all aliquots of medium and filter. Make 50 ml aliquots and store at -80 °C.
- **5.** Supplement CM with additional 4 ng/ml of FGF2 before adding to hESCs grown on Matrigel.

Measurement of Activin A in Conditioned Media (ELISA)

- The cytokine, Activin A, is considered as the most critical factor secreted by feeder cells for the support of undifferentiated growth of pluripotent cells. The measurement of the level of Activin A in CM is a very convenient quantitative assay to monitor the quality of MEFs.



- As the function of the conditioning process is Activin A secretion into the medium by MEFs,
- Activin A is almost undetectable in UM.